Segmentation & Classification Pipeline Comparison Report

U-Net vs. DeepLabV3-ResNet50 - Leaf Growth-Stage Analysis

This report consolidates all evaluation results produced for the U-Net (this repository) and DeepLabV3-ResNet50 (Resnet50test, sibling project) segmentation models, covering segmentation-only performance, growth-stage breakdowns, 5-fold cross-validation, and the full segment→mask→classify pipeline benchmark on the identical 160-image test set.

Section A - Segmentation-Only Performance

Table 1. Train / Validation / Test - Native Training-Loop Metrics

Each model's own training loop logs different metrics: U-Net logs Dice/IoU; DeepLabV3 logs pixel accuracy and IoU. Pixel accuracy is therefore not directly available for U-Net's main run from this source.

Split	U-Net (best ckpt = epoch 58/60)	DeepLabV3-ResNet50 (best ckpt = epoch 15/15, COCO-pretrained backbone)
Train	Dice 0.7974, IoU 0.7187	Pixel-acc 0.9440, IoU 0.7971
Validation	Dice 0.8132, IoU 0.7247	Pixel-acc 0.9365, IoU 0.7963
Test	Dice 0.7572, IoU 0.6650	Pixel-acc 0.9022, IoU 0.7548

Both models generalize similarly from train/val to test (some drop in both cases). DeepLabV3 reaches a higher raw IoU on its own training setup, largely because it starts from a COCO-pretrained backbone (41.99M trainable parameters) versus U-Net's from-scratch 7.77M parameters.

Table 2. Validation / Test - Matched Evaluation Protocol

Both checkpoints (unet_best.pt and deeplabv3_leaf_seg_best.pt) were loaded and run on the real validation/test splits using the same evaluation script, producing per-image Dice, IoU, pixel accuracy, precision, recall, and F1 for both models on equal footing.

Set	U-Net — Dice	U-Net — IoU	DeepLabV3 — Dice	DeepLabV3 — IoU
Validation	0.8132	0.7276	0.7764	0.6433
Test	0.7572	0.6650	0.7536	0.6204

Note on discrepancy: *the DeepLabV3 IoU figures in Table 2 (Val 0.6433 / Test 0.6204) are noticeably lower than the IoU figures logged natively during DeepLabV3's own training in Table 1 (Val 0.7963 / Test 0.7548), even though the Dice/pixel-accuracy figures are closer. U-Net's numbers are consistent between both tables aside from a minor validation-IoU rounding difference (0.7247 vs 0.7276). This gap is worth flagging in the submission — it suggests DeepLabV3's native training-log IoU and the matched-protocol re-evaluation IoU may be computed differently (e.g. averaging method, thresholding, or class handling), and should be double-checked before being used as the headline comparison figure.*

Table 3. Performance Across Growth Stages (Test Set)

Growth Stage	Test Images	U-Net Dice	U-Net IoU	DeepLabV3 Dice	DeepLabV3 IoU
Seedling	38	0.8575 ± 0.068	0.7756 ± 0.091	0.7996 ± 0.065	0.6706 ± 0.084
Developing	41	0.7736 ± 0.152	0.6868 ± 0.169	0.7936 ± 0.179	0.6840 ± 0.179
Flowering	60	0.7289 ± 0.105	0.6313 ± 0.119	0.7353 ± 0.106	0.5906 ± 0.111
Fruiting	21	0.6250 ± 0.212	0.5186 ± 0.200	0.6444 ± 0.158	0.4904 ± 0.131

Both models are weakest on the fruiting stage (fewest test images, most visual variability) and strongest on seedling. DeepLabV3's fruiting-stage Dice/IoU (0.6444 / 0.4904) is nominally higher than U-Net's (0.6250 / 0.5186 IoU) on Dice but lower on IoU — the two metrics disagree at this stage, so it should not be read as a clean win for either model without checking per-image variance (± values are large here for both).

Table 4. Overall Segmentation Performance

Metric	U-Net	DeepLabV3
Mean Pixel Accuracy	0.9280 ± 0.0054	0.9007 ± 0.0650
Leaf Precision	0.7288 ± 0.0335	0.7528 ± 0.1509
Leaf Recall	0.8258 ± 0.0302	0.7798 ± 0.1371
Leaf F1-score	0.7732 ± 0.0081	0.7536 ± 0.1389

U-Net leads on mean pixel accuracy and F1-score; DeepLabV3 leads on precision but has a much wider standard deviation across all four metrics, indicating less consistent performance image-to-image.

Table 5. 5-Fold Cross-Validation - Segmentation Only (Pixel-Level Background/Plant)

Metric	U-Net	DeepLabV3
Test accuracy	0.9280 ± 0.0054	0.9119 ± 0.0050
Leaf/plant precision	0.7288 ± 0.0335	0.8527 ± 0.0049 (macro, both classes)
Leaf/plant recall	0.8258 ± 0.0302	0.8449 ± 0.0154 (macro)
Leaf/plant F1	0.7732 ± 0.0081	0.8485 ± 0.0086 (macro)

Caution: U-Net's CV precision/recall/F1 are for the single "leaf" class only, while DeepLabV3's are macro-averaged over both background and plant classes — so they are not perfectly apples-to-apples (DeepLabV3's numbers get a boost from the easy background class). Test accuracy (U-Net 0.928 vs DeepLabV3 0.912) is the more directly comparable figure and favors U-Net slightly.

Section B - Full Pipeline (Segment → Mask → Classify)

This is the qualitative, apples-to-apples comparison: three segmenter variants feeding an identical, fixed ResNet50 downstream classifier, evaluated on the same 160-image test set.

Table 6. Full Pipeline Comparison — Three Segmenter Variants

Segmenter	Foreground IoU	Foreground Dice	Both-Class Mean IoU	Full Pipeline Test Accuracy
DeepLabV3-ResNet50	0.6230	0.7565	0.7507	0.8438
U-Net (native, direct-resize preprocessing)	0.6646	0.7844	0.7926	0.7875
U-Net (crop-aligned preprocessing) ✓best	0.6674	0.7827	0.7753	0.8938

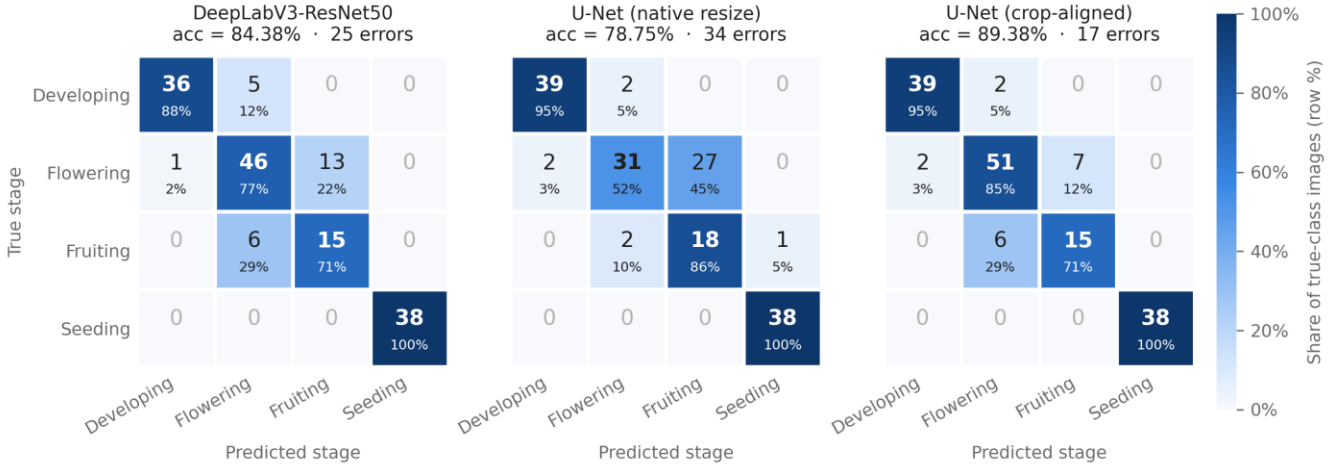
Table 7. Per-Class Pipeline F1

Class	DeepLabV3	U-Net (native)	U-Net (crop-aligned)
Developing	0.9231	0.9512	0.9500
Flowering	0.7863	0.6526	0.8600
Fruiting	0.6122	0.5455	0.7000
Seeding	1.0000	0.9870	1.0000

Key qualitative finding: U-Net's segmentation quality was already at or above DeepLabV3's, but its native preprocessing (direct resize) misaligned the predicted mask geometry with what the classifier expects (Resize(256) → CenterCrop(224)), which reduced pipeline accuracy to 78.75%. Fixing only the preprocessing — crop-aligning U-Net's inference, without touching the segmenter itself — pushed U-Net past DeepLabV3 to 89.38%, the best of all three variants, while segmentation IoU/Dice stayed essentially flat. This indicates the earlier gap was a geometry/alignment bug, not a genuine segmentation-quality gap.

Figure 1. Full-Pipeline Confusion Matrices (Test Set, n = 160)

Rows are true growth stages and columns are predicted stages for the three Table 6 pipelines. Each cell shows the image count and its share of the true class (row %); colour encodes the row %, so classes of different size (21 fruiting vs 60 flowering) are directly comparable. Diagonal cells are correct predictions.



What the confusion matrices show: Seeding is classified perfectly by all three pipelines, and most errors sit in the flowering/fruiting block (19 of 25 for DeepLabV3, 29 of 34 for native U-Net, 13 of 17 for crop-aligned U-Net). Native U-Net misclassifies 27 of 60 flowering images (45%) as fruiting, which accounts for most of its 34 errors. Crop-aligned preprocessing cuts that to 7, and the flowering recall rises from 52% to 85% without retraining the segmenter. This is consistent with the geometry-misalignment explanation above. DeepLabV3 sits between the two (13 flowering→fruiting, 6 fruiting→flowering) and is the only pipeline with notable developing→flowering confusion (5 of 41). Fruiting is the hardest class for every pipeline: 6 of 21 (crop-aligned U-Net and DeepLabV3) are predicted as flowering, which matches its lowest F1 in Table 7.

Provenance: U-Net (crop-aligned) counts are recomputed from the per-image predictions (full_pipeline_test_results.csv); U-Net (native) counts are from the saved comparison-notebook confusion matrix. The DeepLabV3 matrix is for the 0.8438 run used in Tables 6–7. It was reconstructed from that run's saved classification report (per-class support, recall and F1), which determines the counts uniquely. A separately saved DeepLabV3 rerun with 0.8750 accuracy has a slightly different matrix and is not shown, so that the figure matches Table 6.

Table 8. 5-Fold Cross-Validation - Full Pipeline

Metric	U-Net-based pipeline (fixed classifier)	DeepLabV3-based pipeline (retrains both stages per fold)
Pipeline accuracy	0.8562 ± 0.0238	Fold 1: 0.9068 · Fold 2: 0.9237 · Fold 3: 0.9237 (partial — full mean not yet captured)
Macro F1	0.8441 ± 0.0237	Fold 1: 0.7916 · Fold 2: 0.7690 (partial)

These two CV setups are not directly comparable: this repository's Part B CV reuses one fixed classifier across all folds (isolating the segmenter's own contribution), while DeepLabV3's §10 CV retrains both the segmenter and the classifier per fold. The DeepLabV3 fold log is partial — folds 4–5 and the full mean were not captured in this pass.

Key Findings

- Segmentation only: U-Net edges out DeepLabV3 on mean pixel accuracy and F1 (Tables 4–5); DeepLabV3 edges out U-Net on raw training-setup IoU, aided by a COCO-pretrained backbone (Table 1).
- Full pipeline: U-Net with crop-aligned preprocessing is the best of all three variants tested, reaching 89.38% pipeline accuracy versus DeepLabV3's 84.38% (Table 6).
- The single biggest lever in this study was not the segmentation model choice but preprocessing alignment between the segmenter's output and the classifier's expected input geometry — fixing that alone raised U-Net's pipeline accuracy by over 10 points (Table 6).
- Both models are consistently weakest on the fruiting growth stage across every table, likely reflecting the smaller sample size (21 test images) and higher visual variability at that stage.
- Cross-validation results (Tables 5 and 8) are directionally consistent with the single-split results but use different comparison protocols on each side and should be read with that caveat rather than as a strict head-to-head.
- Confusion matrices (Figure 1): most pipeline errors are flowering↔fruiting confusions. Crop alignment reduces U-Net's flowering→fruiting errors from 27 to 7, which accounts for most of its accuracy gain, and seeding is perfectly classified by every variant.